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Sounding Rocket Structural Design Loads (rev.9)

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Abstract

This report documents the math models used in BENDIT7.xls, a code for developing structural loads acting on sounding rockets. Body shear force, body bending moment and fin bending moment are reported for input flight conditions and probability of success. Limited corrections for body static aeroelasticity are incorporated. Either one or two stages, each with fins can be considered. Gauss/Rayleigh statistics for wind gusts, thrust misalignment, fin misalignment and center of mass offset are reflected in these estimates. Three possible roll rate profiles can be used. Mean body axial loads due to thrust and parachute deployment are calculated. Body ground handling loads are estimated, Finally, as a by product, rocket mass properties and stability derivatives are reported.

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Introduction

The process of designing the structure of a new sounding rocket, or an element thereof, requires estimates of the structural loads imposed and the probability they will be exceeded, i.e., the risk of structural failure This is true for both fins and body of the rocket. Numerical results may be found from an associated spreadsheet, BENDIT, while this memo supplies the underlying rationale and derivations used to build BENDIT. BENDIT assumes rigid body motion with no pitch/yaw damping[1]. For the body and its fins four perturbing effects are important, thrust misalignment, fin misalignment, and wind gusts. Thrust misalignment includes both the effects of the body CG offset from its symmetry axis and angular misorientation between the thrust vector and the body axis of symmetry. Gaussian statistics are used throughout.

The loads described in this memo are only those arising from external forces and aerodynamics. Loads arising from any pressure difference between the external environment and internal cavity pressures are assessed elsewhere.

Nomenclature

Mnemonic	Definition
A_x	Burnout axial acceleration, ft/sec^2
A_z	Normal acceleration due to angle of attack, ft/sec^2
b	Fin semispan , measured from the rocket centerline, ft
B	Exposed fin semispan, ft
$c(y)$	Fin local chord at spanwise station y, ft
c_R	Root chord of an exposed fin panel, ft
c_T	Tip chord of a fin panel, ft
C_{DP}	Parachute drag coefficient
C_{Di}	Drag coefficient of the ith element
C_L	Rolling moment coefficient ≡ Rolling moment / $qS_{ref}d_{ref}$
C_{LP}	Rolling moment coefficient due to roll rate, $(\omega_R R/U)^{-1}$ for the entire rocket
$C_{L\delta F}$	Roll driving stability derivative for a single fin panel, 1/rad
C_M	Pitching moment coefficient ≡ Pitching moment / $qS_{ref}d_{ref}$
$C_{M\alpha}$	Aerodynamic pitching moment coefficient due to angle of attack, rad^{-1}
$C_{M\delta F}$	Single panel aerodynamic pitching moment coefficient due to a cant angle, rad^{-1}

Mnemonic Definition

[1] Pitch (and yaw) damping arises from three distinct terms in the equations of motion: first, aerodynamic pitch damping (C_{Mq}), second, aerodynamic plunging motion damping ($C_{z\alpha}$), and third, rocket motor jet damping, (M_J). Acting together, they damp the classical short period mode to much less than 1% of critical.

C_N	Normal force coefficient ≡ Normal force / qS_{ref}
$C_{N\alpha}$	Normal force coefficient slope for the entire rocket, rad^{-1}
$c_{N\alpha}$	Average isolated single panel two dimensional fin airfoil normal force coefficient slope including tip effects, rad^{-1}
$C_{N\alpha i}$	Normal force coefficient slope of the ith element, rad^{-}
$C_{N\alpha N}$	Normal force coefficient slope of the upper stage/dart nose tip, rad^{-1}
d_{ref}	Aerodynamic reference length, ft
D_P	Drag deceleration at parachute deployment, ft^2/sec
DL	Design load (either shear force or bending moment)
EI	Bending stiffness of the upper stage/dart, lbft^2
F	Nose tip normal force amplification due to flexibility
h	Altitude above ground level, ft
I_{YY}	Total rocket pitch / yaw moment of inertia, sl-ft^2
J_i	Moment of inertia about the center of mass of the ith element, sl-ft^2
$k_i l_i$	Center of mass from the front of the ith element, ft
$K_{W(B)}$	Normal force on a set of fins at angle of attack in the presence of the body / normal force acting on the fins at angle of attack in isolation
$K_{B(W)}$	Normal force on the body at angle of attack in the presence of the fins / normal force acting on the fins at angle of attack in isolation
$k_{W(B)}$	Normal force on a fin at cant angle in the presence of the body / normal force acting on a fin at cant angle in isolation
$k_{B(W)}$	Normal force on the body due to a fin at cant angle in the presence of the body / normal force acting on a fin at cant angle in isolation
L	Launcher length, ft
l	Axial distance from nozzle throat to rocket center of mass, ft
l_i	Length of the ith element, ft
l_N	Length of the upper stage/dart from the nose tip center of pressure to the base of the upper stage/dart, ft
L_{fin}	Fin bending moment, ft-lb
M_i	Bending moment acting on the forward face of the ith element, ft-lb
$M(y)$	Fin bending moment at spanwise station y, ft-lb
m	Mass of the entire rocket, sl
m_i	Mass of the ith element, sl

Mnemonic	Definition
N	Number of fin panels
P_i	Axial force, positive in compression, acting on the front face

	of the ith element, lb
Pr	The probability that the design load DL will not be exceeded.
q	Dynamic pressure = ½ ρU^2, lb/ft^2
q_D	Nose divergence dynamic pressure, lb/ft^2
q_P	Dynamic pressure at parachute deployment
R_L	Radius from the body center line to the area centroid of a longeron, ft
R_i	Mean element radius (Average over element length), ft
S_i	Shear force acting on the forward face of the ith element, lb
S_P	Parachute drag area, ft^2
S_{ref}	Aerodynamic reference area, ft^2
T	Liftoff thrust, lb
TL	Fin tensile load, lb
t	Time after liftoff, sec
U	Flight speed, ft/sec
$\bar{X}$	Total rigid body center of mass from the nose tip, ft
xCM	Distance from the fin root chord leading edge to the $\bar{X}$ center of mass, ft
x_i	Distance from the nose tip to the forward boundary of the ith element, ft
x_{cgi}	Center of mass of the ith element from the nose tip, ft
x_{cpi}	Center of pressure of the ith element from the nose tip, ft
$x_{CPF\alpha}$	Center of pressure of a fin set in the presence of the body due to angle of attack, in
$x_{CPB\alpha}$	Center of pressure of the body in the presence of a fin set due to angle of attack, in
x_F	The distance from the nose tip to the exposed fin panel root chord leading edge, ft
y	Spanwise distance from the rocket center line, ft
α	Angle of attack, rad
Γ	Fin leading edge sweepback angle, rad
$\Delta C_{M\alpha}$	Increment in $C_{M\alpha}$ due to the influence of trailing second stage fin vortices impinging on the first stage fins, rad^{-1}
$C_{N\alpha V}$	Increment in first stage fin normal force coefficient due to the influence of trailing second stage fin vortices on the first stage fins, rad^{-1}
δ_F	Single panel fin misalignment angle, rad

Mnemonic	Definition
δ_T	1 σ, 1 plane thrust misalignment angle
ϕ	Total roll angle since launcher departure, rad
λ_G^*	Reciprocal of 0.59*the longitudinal gust correlation scale length, ft^{-1}

λ_P	Pitch/yaw wavenumber, radians/foot of altitude
λ_R	Roll wavenumber, radians/foot of altitude
μ	Fin panel mass density per unit area, sl/ft^2
ρ	Atmospheric mass density, sl/ft^3
σ	The planar standard deviation at a specific body station of the shear force or bending moment
σ_G	1 σ, 1 plane gust velocity, ft/sec
$\sigma_{\delta F}$	Standard deviation in fin cant angle for a single fin panel, rad
Ω	Rocket center of mass offset from the axis of symmetry, ft
ω_N	Short period pitch/yaw natural frequency, rad/sec
ω_R	Roll rate, positive according right hand rule with thumb pointing in the flight direction, rad/sec
$Avg()$	The ensemble averaging operator acting on **( )**
$Var()$	The variance operator acting on **( )** = differencing from mean, squaring & ensemble averaging,

Perturbations

I. Thrust Misalignment

General experience has shown that misalignment of the thrust vector with respect the rocket's center of mass is the most important of the various body-fixed misalignments arising from the fabrication and assembly processes used to create rocket hardware. Since the angle of attack contributions from these are usually independent, the overall standard deviation in angle of attack is the RSS of the contributions of each. In this case, only the dominant effect (thrust misalignment) has significant influence. Note that thrust misalignment has two independent, orthogonal components in pitch and yaw.

Reference (1) documents the sounding rocket angle of attack response history due to a thrust vector misalignment. It assumes that pitch/yaw damping is negligible, and finds the angle of attack response by solving the 4 DOF axisymmetric, time-varying equations of motion assuming constant axial acceleration. By changing independent variable from time to altitude, the equations are converted to new ones with constant coefficients. Since the initial angle of attack is zero as the rocket separates from its launcher, it should come as no surprise that the angle of attack initially overshoots the steady state solution and oscillates. As dynamic pressure increases, the oscillation amplitude decreases. For loads analysis, the oscillation amplitude envelope should be used because accounting for the details of phasing is more trouble than it's worth.

II. Wind Gusts

Wind gusts are stochastic turbulence in the atmosphere. See reference (2). Unlike synoptic scale winds, the main issue with gusts is that they cannot be deterministically predicted for more than a small fraction of a second ahead.

This immediately precludes any predictive scheme for mitigating the gust response. The variance in angle of attack at any altitude is found by summing over all the lower altitude bands accounting for the Batchelor correlations between altitudes. Gusts are modeled as having a Dryden autocorrelation function with Gaussian statistics. The angle of attack impulse response function is obtained, assuming constant axial acceleration, after changing the independent variable from time to altitude. Closed form integrals for the gust angle of attack variance are found.

III. Fin Misalignment

This is basically a fin panel angular orientation issue. Many fins are assembled to their rocket bodies without much care for a desired orientation. Best practice requires each fin to be surveyed in cant angle at several spanwise stations, and an angular cant angle adjustment at its root made to minimize pitch/yaw misalignment torques while achieving the desired roll rate. While fin misalignments also cause angle of attack excursions, in practice these are sometimes found to be of significantly lesser importance than those due to thrust misalignments or gusts.

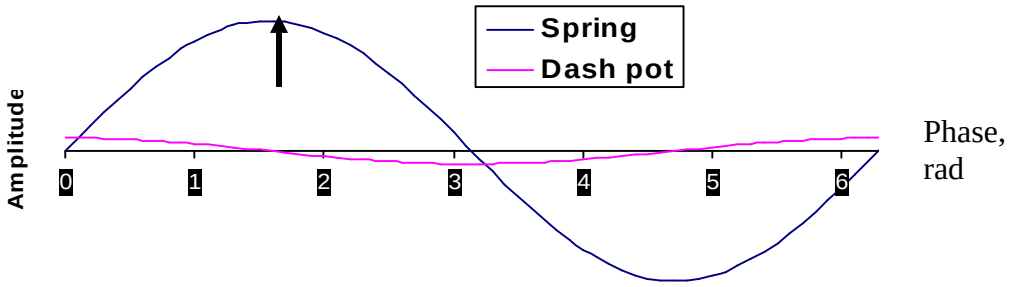
Solving the roll equation of motion is commonly done by assuming a quasi-static condition in which the roll moment of inertia is neglected. However, BENDIT7 also estimates the initial roll rate lag immediately after burnout. A key assumption is that any roll moments induced on the fin panels by vortices shed from the forebody at high angle of attack are not considered here.

Loading Conditions

The two principal sources of transverse loading, thrust misalignment and wind gusts have similar consequences. Thrust misalignment, see ref. (1), generates an oscillatory response that is essentially undamped. Gusts are stochastic and aperiodic, but the rocket's small short period damping makes the angle of attack power spectral density show a pronounced spike at the fundamental natural frequency/wave number.

We assume therefore that the rocket is oscillating in, say pitch, and the maximum air loads occur when the instantaneous angle of attack is at its peak value. Furthermore, we

of



mass[2]. However, when there is a non-zero angle of attack, aerodynamic normal forces



Relative Loading

mass[2]. However, when there is a non-zero angle of attack, aerodynamic normal forces will cause the center of mass to accelerate laterally. It's just that this acceleration results in negligible short period plunging damping.

The sketch above displays the relative amplitude of the aerodynamic spring / inertia torques and of the dash pot, or damping torque. The damping shown has been exaggerated to make the point; the true level is about10% of that shown. The maximum loads are indicated by the arrows.

For the tail fins, the angle of attack oscillation still occurs. But the local angle of attack on a tail fin also has contributions from the fin root cant angle and the roll rate. The fin loading condition is a steady state one to which the oscillatory angle of attack is added.

The next issue is where in flight one expects he maximum loads to occur. Selection of events to be analyzed for design loads is the responsibility of the Loads Analyst whose judgment is essential. Maximum dynamic pressure is an obvious example of a candidate loads analysis event. Others are stage ignition, burnout & parachute deployment. Because the dominant body-fixed perturbations are perfectly correlated over time, the loads at different flight epochs are also correlated. Thus, the local maximum loading at, say maximum dynamic pressure, is also approximately the global maximum loading.

BENDIT7 is designed to only analyze one condition at a time. For example, the parachute deployment loads for drogue and main chute deployment require two separate, distinct analyses even though much of the data used is the same for both. For a single stage rocket the maximum loading usually occurs near burnout. For a two stage rocket, there are typically three possible maximum loading flight conditions, near first stage burnout, second stage ignition and near second stage burnout. Local maximum dynamic pressures will occur near burnout.

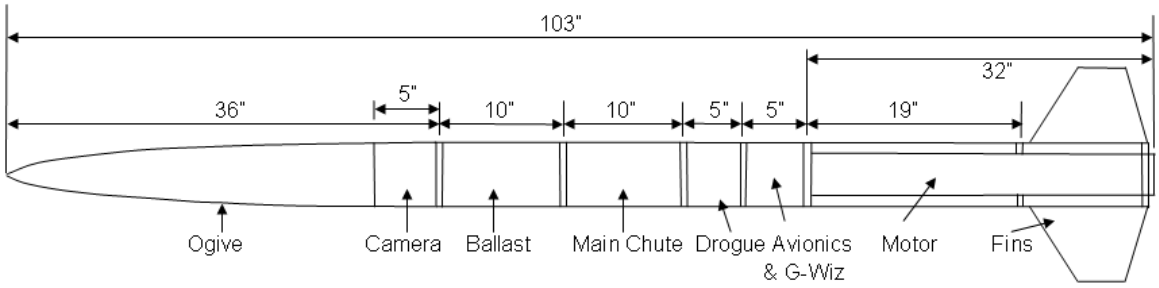
Body Elements

Begin by constructing an inboard profile of the rocket. This shows the internal configuration of the rocket with dimensions. Since many rockets are built with a longeron and bulkhead structural concept, it's convenient to partition the internal bays

[2] Freeing the rocket to plunge would only add a negligibly small amount of short period damping. In other words, the difference between angle of attack and inertial pitch angle is also negligible.

along the bulkhead midplanes. Consider the notional inboard profile below of an ESRA Basic Category rocket, together with other dimensional data needed:

- Body outside diameter = 6.0”
- Fin root chord = 10”
- Fin tip chord = 4”
- Exposed semispan = 8”
- Fin thickness = 0.125”
- Nose profile is a 6 to 1 tangent ogive



Typical Inboard Profile of an ESRA Sounding Rocket

Now decompose the body into elements. Good practice considers an element to be everything between the midplanes of a pair of adjacent bulkheads. Everything includes skin & paint, longerons, functional equipment, wiring & fasteners and all, or part, of bulkheads. But, an element should not be too long; anything over about two body diameters. Excessively long elements should be broken up into multiple short elements. Finally, it’s good practice to select element boundaries such that the entire fin root chord is contained within the boundaries of a single element.

Each element is assumed to have uniform mass density. Then the mass properties of an element are

Length $= x_{i+1} - x_i = l_i$,

Mass $= m_i$[3],

Center of mass from the front of an element $= k_i$[4] $l_i = x_{cgi}$,(uniform density), and

The air load acting on the element is: $= qS_{ref} C_{N\alpha i} \alpha$

Moment of inertia about element center of mass $= \left(\frac{m_i}{12}\right)\left(3R_i^2 + l_i^2\right) = J_i$.

The aerodynamic torque (+ right hand rule, thumb into the paper) is:

Aerodynamic torque[5] (+ right hand rule, thumb into the paper)

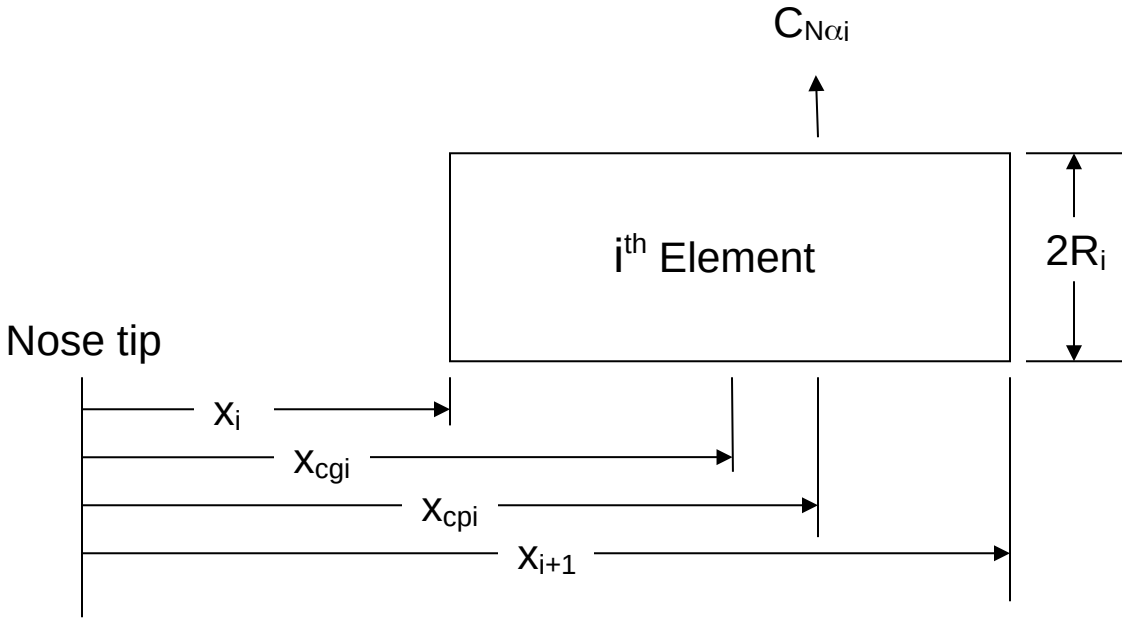
[3] These formulas are not valid for the body element to which the fins are attached. See Appendix A for a detailed description of the mass properties of this element.

[4] See Appendix B.

[5] The body elements carrying fins are assumed to have constant radius, and hence will have nearly zero subsonic $C_{N\alpha i}$ according to slender body theory. Even at supersonic speeds their normal force will be small. The $C_{N\alpha i}$ is essentially all due to the fins, fin-

$$= qS_{ref} C_{N\alpha i} \alpha (x_{cgi} - x_{cp[i}).$$

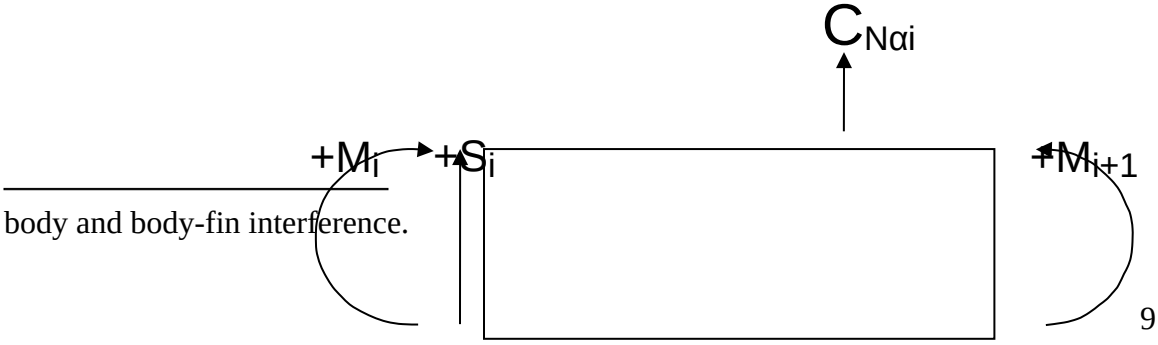
Finally, note that the implementation of this memo is BENDIT7.xls. This code, originally intended only for loads estimation, can also be used to estimate the rocket's mass properties.



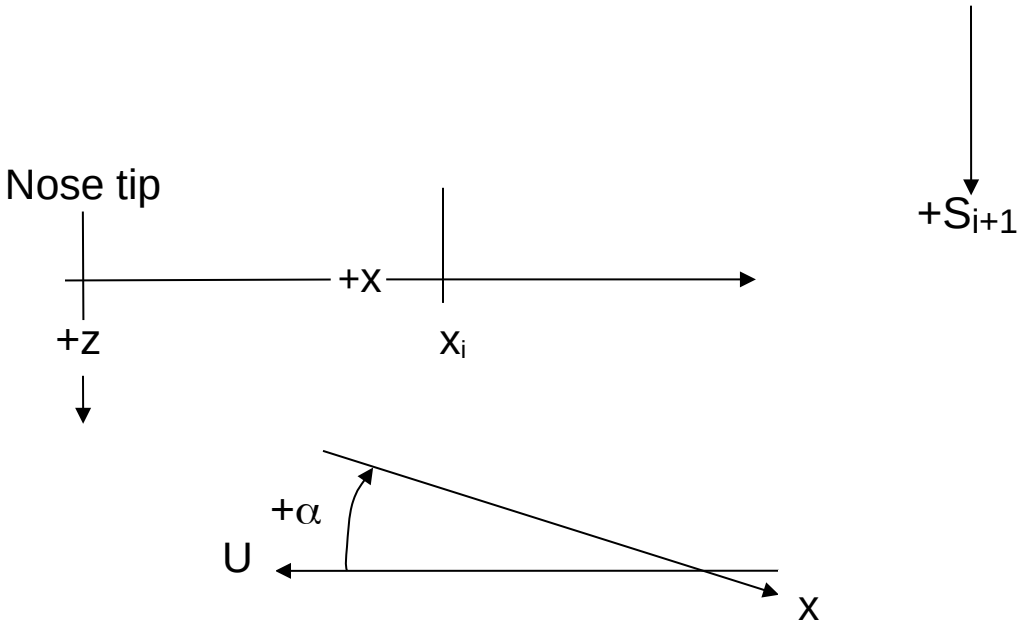
Body Element Dimension Definitions

Difference Equations and Initial Conditions

Now, apply Newton's second law to an element, in both the force and torque senses. The sketches below are guides that establish the relevant sign conventions and the coordinate frame used.



body and body-fin interference.



Element Sign Conventions

Summing the forces acting on the element in the +z direction gives

$$S_{i+1} - S_i - qS_{ref} C_{N\alpha i}\alpha = m_i \left(A_z - (X - x_{cgi}) \frac{d^2\alpha}{dt^2} \right), \text{ where} \qquad (1)$$

X = The total rigid body center of mass given by

$$X = \frac{\sum m_i x_{cgi}}{\sum m_i}, \text{ and} \qquad (2a)$$

A_z = The normal acceleration given by

$$A_Z = - \frac{qS_{ref}\alpha \sum C_{N\alpha i}}{\sum m_i} \qquad (3)$$

Similarly, summing the y axis torques (right hand rule) about the element center of mass gives

$$M_i - M_{i+1} + (x_{cgi} - x_i) S_i + (l_i + x_i - x_{cgi}) S_{i+1} + qS_{ref} C_{N\alpha i}\alpha (x_{cgi} - x_{cpi}) = J_i \frac{d^2\alpha}{dt^2} \qquad (4)$$

The nose tip initial conditions are that

$$S_1 = M_1 = 0 \tag{5}$$

A marching solution of these difference equations is straightforward. Starting with the known value of S_1, eq. (1) can be used to find S_2. Then, given M_1, S_1 and S_2, eq. (4) can be used to find M_2. This marching process can be used for the second element, and so on.

Finally, it must be emphasized that the air load acting on a body element is composed of contributions from both the body itself and any attached fins. For a discussion of the latter, see Appendix F.

Rigid Body Motion

Equations (1) and (4) have terms in A_z, α and $\frac{d^2\alpha}{dt^2}$. The rigid body moment equation relates the latter two. First, the total moment of inertia can be found from the parallel axis theorem:

$$I_{YY} = \sum J_i + m_i (X - x_{cgi})^2 . \tag{2b}$$

Then,

$$I_{YY} \frac{d^2\alpha}{dt^2} = qS_{ref} \alpha \sum C_{N\alpha i} (X - x_{cpi}). \tag{6}$$

As noted in the Introduction, α can be one of the major BENDIT inputs, or one can estimate it in BENDIT from first principles as described in Appendix D.. Using eq's (3) and (6) to find A_z and $\frac{d^2\alpha}{dt^2}$ enables the marching solution of eq's. (1) & (4).

Static Aeroelastic Effects

Up until now, the rocket body has been assumed to be perfectly rigid. But, sometimes rockets have slender upper stages or darts much more susceptible to bending than the main vehicle. At angle of attack their nose curls up increasing the normal force loading. And, the increased normal force causes additional deflection, and so on. In theory this cycle of air load and deflection can continue until divergence occurs. In practice this cannot happen because the increased loads cause structural failure first.

Our point of departure is equations (1), (3), (4) and (6). The way to look at this problem is to consider that the $C_{N\alpha}$ for the nose is increased by a factor F to allow for flexibility effects. Equations (3) and (6) show that the rigid body motions, A_z and $\frac{d^2\alpha}{dt^2}$, are slightly affected by the increase in nose normal force. Then, to estimate F model the flexible

nose as a uniform beam with a concentrated load at its tip. Reference (10) describes this process in detail. The slope $\frac{dy}{dx}$ of the bent nose at its tip is

$$\frac{dy}{dx} = (qS_{ref} C_{N\alpha N}(\alpha + \frac{dy}{dx}) + m_N(A_z - (X - x_{CGN})\frac{d^2\alpha}{dt^2}))l_N^2 / 2EI . \qquad (7a)$$

After substituting from eq's. (3 and 6), this can be solved for $\frac{dy}{dx}$, and then for the total nose tip angle of attack,

$$\alpha_{TOT} = \alpha + \frac{dy}{dx} = \alpha \left[\frac{2EI - qS_{ref}\left[\frac{m_N}{m} C_{N\alpha} + \frac{m_N(X - x_{CGN})}{I_{YY}} \sum C_{N\alpha i}(X - x_{CPi}) \right] l_N^2}{2EI - qS_{ref} C_{N\alpha N} l_N^2} \right] \qquad (7b)$$

Note that the dynamic pressure which will make the denominator vanish in eq. (7b) is the divergence dynamic pressure:

$$q_D = \frac{2EI}{S_{ref} C_{N\alpha N} l_N^2}$$

Next, to incorporate this effect in BENDIT7, it is only necessary to increase the nose tip normal force slope to

$$C_{N\alpha N} \left[\frac{2EI - qS_{ref}\left[\frac{m_N}{m} C_{N\alpha} + \frac{m_N(X - x_{CGN})}{I_{YY}} \sum C_{N\alpha i}(X - x_{CPi}) \right] l_N^2}{2EI - qS_{ref} C_{N\alpha N} l_N^2} \right] = F C_{N\alpha N} .$$

(7c)

That is, in eq. (7c) the term in brackets is the amplification factor, F, applied to all elements forward of x_N .

When eq. (7c) is substituted into eq's. (3) and (6) the result is we still have two linear equations with two unknowns giving solutions for A_z and $\frac{d^2\alpha}{dt^2}$, both proportional to α.

Finally, note that the bending stiffness is the product of the material Young's modulus, E, and the cross section moment of inertia, I. For a beam of circular cross section,

$$I = \frac{\pi}{4}(R_o^4 - R_i^4),$$

where R_o and R_i are the outer and inner radii respectively.

Axial Load

It's time for a cautionary note. In many rocket bodies parallel beam structures can be found. Examples are a payload rack inside a fairing or a rocket engine, surrounded by fins and aft fairing, attached at its injector to the main structure. Good design practice suggests these internal beams be laterally supported by the main structure over their length. That is, a good approximation is that such internal beams carry only axial load. The analysis reported here should be all right so long as the mass distributions for lateral bending and axial loading be distinguished and analyzed separately.

The axial loads acting on an accelerating rocket are compressive forward of the body station at which the thrust is applied, and tensile at more after body stations. When thrust is applied to the after part of the rocket the inertial reaction comes from the masses above/forward. Experience indicates that burn out is the most severe steady loading condition. Then, the burn out acceleration is

$$A_x = (T - qS_{ref} \Sigma C_{Di}) / \Sigma m_i \,. \tag{8a}$$

Next, assume the thrust is applied to the aft face of the tth (pronounced "teeth") element. Then, the compressive load acting on the front face of the 1st element must vanish $(P_1 = 0)$ while the compressive load acting on the front face of the 2nd element is $m_1 A_x + qS_{ref} C_{D1}$. The general recursion formula needed for a solution marching down towards the tth element is

$$P_{i+1} = P_i + m_i A_x + qS_{ref} C_{Di} \tag{8b}$$

Below/aft of the tth element the axial load is tensile. Marching forward from the aft end where the axial load also vanishes results in a similar recursion relationship:

$$P_i = P_{i+1} - m_i A_x - qS_{ref} C_{Di} \tag{8c}$$

Note that tensile load carries a negative sign. Equations (8a) and (8b) can be integrated by marching just like eqs. (1) and (4). Modification of these equations to account for parallel structures, such as a payload rack inside a payload fairing, is straightforward.

Finally, the above results are for statically loading the structure. In fact, the rocket structure should be considered to be an elastic column. At ignition, this column is subjected to rapidly rising compressive load from its propulsion system. This generates an elastic wave that propagates up the "stack". The result is that in many parts of the structure the instantaneous axial load is several times the steady state value. Rather than attempting a detailed analysis of this process, a simple amplification factor is applied to

all axial loads at an ignition event. The simplified model in ref. (22) suggests that a factor of 3 be used for this purpose.

Fins

The fins are decomposed into chordwise elements in a way directly analogous to what was done for the body. Consider the sketch below of a fin airfoil used to define sign conventions at an arbitrary spanwise station:

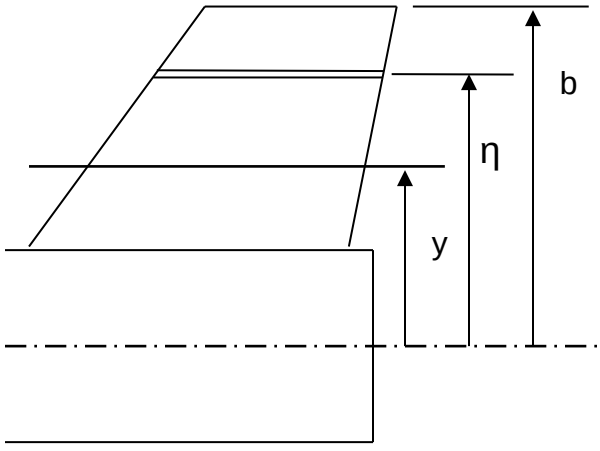


Fin Angle of Attack Sign Convention

The local angle of attack experienced by an element of fin airfoil is

$$\alpha local = \alpha\left[1+(R/y)^2\right] + \delta_F - \omega_R y/U + \sum_1^N v/U = \alpha\left[1+(R/y)^2\right] + \delta_F - \lambda_R y + \sum_1^N v/U \quad (9)$$

Here, the body angle of attack α has been corrected per ref. (4) for incompressible cross flow upwash. In the case of a first stage fin, the downwash field from the second stage fins gives rise to the v/U terms. For an ordinary four-finned configuration this is modeled by 4 discrete vortices. See ref. (21) for a complete description of this vortex field, its angle of attack distribution and the resulting bending moments.



Fin Bending Moment Geometry

Using strip theory, one can readily integrate the roll torque / aerodynamic bending moment resulting from this local angle of attack. See ref. (3) for a more detailed discussion of this topic.

$$M(y) = \int_{y}^{b} qc(\eta) c_{N\alpha F} \alpha.local(\eta - y) d\eta.$$

(10a)

The local chord is given by

$$c(\eta) = A - C\eta = (c_R b - c_T R)/(b - R) - (c_R - c_T)\eta/(b - R)$$

(10b)

The fin bending moment is found by substituting eq's. (9) and (10b) into eq. (10a), and integrating. The result is that

$$\begin{aligned} &M(y)/qC_{N\alpha F} = \\ &\alpha\left[-Ay((b-y) + R^2(b-y)/by) + (A+Cy)((b^2-y^2)/2 + R^2\log(b/y)) - C((b^3-y^3)/3 + R^2(b-y))\right] \\ &+ \delta_F\left[-Ay(b-y) + (A+Cy)(b^2-y^2)/2 - C(b^3-y^3)/3\right] \\ &+ \lambda_R\left[Ay(b^2-y^2)/2 - (A+Cy)(b^3-y^3)/3 + C(b^4-y^4)/4\right] \\ &+ \alpha\sum_{1}^{N} V(y) \end{aligned} \qquad (10c)$$

The last term, $\sum_{1}^{N} V(y)$, is the bending moment on first stage fins induced by the vortices shed from the second stage fins.

In going forward it will be convenient to define new functions based on eq. (10c):

$$M(y) = qC_{N\alpha F}\left[\alpha F(y) + \delta_F G(y) + \lambda_R H(y)\right]$$

(11a)

Using a nominal *mean* values of δ_F and λ_R, one can easily find the nominal *mean* spanwise distribution of bending moment from eq. (11a), noting that the mean value of α is zero. Note also that the roll inertial reaction moments have been neglected here because the rocket's roll acceleration is small. But, the pitch/yaw inertial reaction loads due angle of attack are not negligible. Replace $F(y)$ in eq. (11a) with $F_{TOT}(y)$ from eq. (19b). See Appendix C for a fuller description of the inertial relief fin bending moment.

Finding the standard deviation of bending moment is a little trickier. First, assume that the fin cant angle errors are all mutually independent, and, also their correlations with α must be included. That is, assume δ_F errors are not independent of α errors. Also, assume that the cant angle error statistics are the same for all stages. Then,

$$Var(M(y)) = q^2 c_{N\alpha F}^2 \begin{bmatrix} Var(\alpha)F^2(y) + Var(\delta_F)G^2(y) + Var(\lambda_R)H^2(y) + 2Avg(\alpha\delta_F)F(y)G(y) \\ + 2Avg(\alpha\lambda_R)F(y)H(y) + 2Avg(\delta_F\lambda_R)G(y)H(y) \end{bmatrix}$$

(12a)

Evaluate each of the terms in eq. (12a) in turn: The variance in α can be found in Appendix D. The variance in δ_F is an input variable. The variance in roll wavenumber λ_R requires consideration of the approximate rigid body steady state roll rate equation,

$$\lambda_R R \sum_{stages} C_{LP} + \sum_{stages} \left[C_{L\delta F} \sum_{1}^{N} \delta_F \right] = 0,$$

(13a)

where the summations are over all the stages in the configuration analyzed. Then, the mean roll wavenumber is

$$\lambda_R = - \frac{\sum_{stages} C_{L\delta F} N \delta_F}{R \sum_{stages} C_{LP}},$$

(13b)

and the variance is

$$Var(\lambda_R) = \frac{N \sum_{stages} C_{L\delta F}^2 Var(\delta_F)}{R^2 C_{LP}^2} \cdot$$

(13c)

Next,

$$2Avg(\alpha\delta_F) = \frac{\rho S_{ref} d_{ref} C_{M\delta F} Var(\delta_F)}{I_{YY}(\lambda_P^2 - \lambda_R^2)},$$

(12b)

and

$$2Avg(\alpha\lambda_R) = 0. \qquad (12c)$$

The cross correlation between α and λ_R vanishes because the α response has two δ_F terms of opposite sign from the fins on opposite sides of the rocket while λ_R has the same two δ_F terms, only with the same sign. Thus the two δ_F variances cancel each other.

Finally,

$$2Avg(\delta_F\lambda_R) = - 2 \frac{C_{L\delta F} Var(\delta_F)}{R \sum_{stages} C_{LP}}$$

(12d)

In the development of eq's. (12) one should keep in mind when ensemble-averaging terms with δ_F that this is just the cant angle of a single fin panel independent of all other cant angle errors. The variance in fin bending moment is found by substituting the above results into eq. (12a). Remember that the inertial reaction bending moment found in Appendix C leads to a modified form of the function F(y).

The strip theory results for C_{L_P} and C_{L_δ} are developed in ref. (5), and reported in Appendix E. As always, $Var(\alpha)$ is the variance sum of the wind gust, and thrust/CG offset and fin misalignment αs.

The last fin loading topic arises from the fact that rolling will load the fins in tension. Although this effect is usually small, it should be included in any fin loading analysis. See Appendix I for the details.

Parachute Deployment

Another important flight condition that should be considered when analyzing structural loads is parachute deployment. This applies to both drogue and main parachute deployments. The CSULB ESRA rockets all store their parachutes in the rocket body forward of the rocket motor. Their risers are laid on the outside of the rocket in faired trays. Attachment to the racket structure is at the aftermost bulkhead.

At deployment a small black powder charge expels the parachute from its storage bay. As it inflates it swings into position directly behind the rocket. The design loading condition happens just as inflation behind the rocket is completed but before significant deceleration can occur. The maximum deceleration at that time is D_P/m. Here D_P is the opening shock as found from OSCALC in ref. (11). For the CSULB ESRA design the resulting structural loading is entirely in tension. It is given by

$$P_i = - D_P \sum m_i ,$$

(14)

where the summation is over all the elements forward of body station i. As a caution to the analyst, note that the mass of the deployed parachute, risers and bridle has been moved from their storage bay to their deployed location behind the rocket.

Next, suppose the parachute shock cord is attached to the forward bulkhead of the Nth element. If $x_N \le X$ it will be assumed that the rocket will instantaneously swing to a tail down orientation when the parachute is deployed. If $x_N \ge X$ it will swing nose down. In the first case the axial force on elements ahead of x_N will be in compression while those aft of x_N will be in tension. In the second case, this is just reversed.

Finally, the transient amplification factor used to correct estimated static axial loads for transient effects applies here as well. A value of three for the amplification factor is recommended.

Ground Handling Loads and Venting

Although these topics are not described in this memo, they can significantly influence the estimated loads. Ground handling loads are documented in ref. (11), while only carrying loads are implemented in BENDIT7. Venting of rocket compartments and bays occurs naturally as a rocket ascends in the atmosphere. If one is not careful to provide proper venting holes, the pressure due to trapped air inside the rocket can easily burst the skin. The venting process is captured in ref. (13). Venting calculations are implemented in BLOWDOWN2.xls.

Combinatorics

The results so far described are the shear force and bending moment distribution along a rocket body for a given α. Furthermore, if the α value used is interpreted as a standard deviation (the root-sum-square of the contributions from thrust misalignment and wind gusts) then the shear force and bending moment at any body station are also standard deviations. Presumably these will follow a normal (Gaussian) distribution with zero mean. But, rocket bodies are rotationally symmetric about their center line. That is, in a plane orthogonal to the α plane (the β plane) there will be exactly the same distribution of shear force and bending moment. Because the α and β orientations are arbitrary, it's the amplitude of the shear force and bending moment that matter, and these, for the above reasons, will follow a Rayleigh distribution:

$$DL = \sigma\sqrt{-2\log(1-\mathrm{Pr})} \text{ , where} \qquad (15)$$

DL = Design Load,
σ = Standard deviation in the single plane load, and
Pr = Probability the load DL will not be exceeded.

As an example, if the loads are to be exceeded no more than once in a thousand flights, eq. (15) gives

$$DL = 3.717\sigma \qquad (16)$$

This offers a powerful technique to manage an important flight risk, that of structural failure.

This result is easy to apply. First, consider a longeron-and-bulkhead structure with L equally spaced longerons. According to ref. (19), the maximum / minimum axial load P_L acting on a longeron is

$$P_L = \pm\frac{2M_B}{RL} \qquad (17)$$

Since the maximum / minimum bending moment M_B has a random roll angular orientation, the worst case longeron axial load needed for design is found by substituting

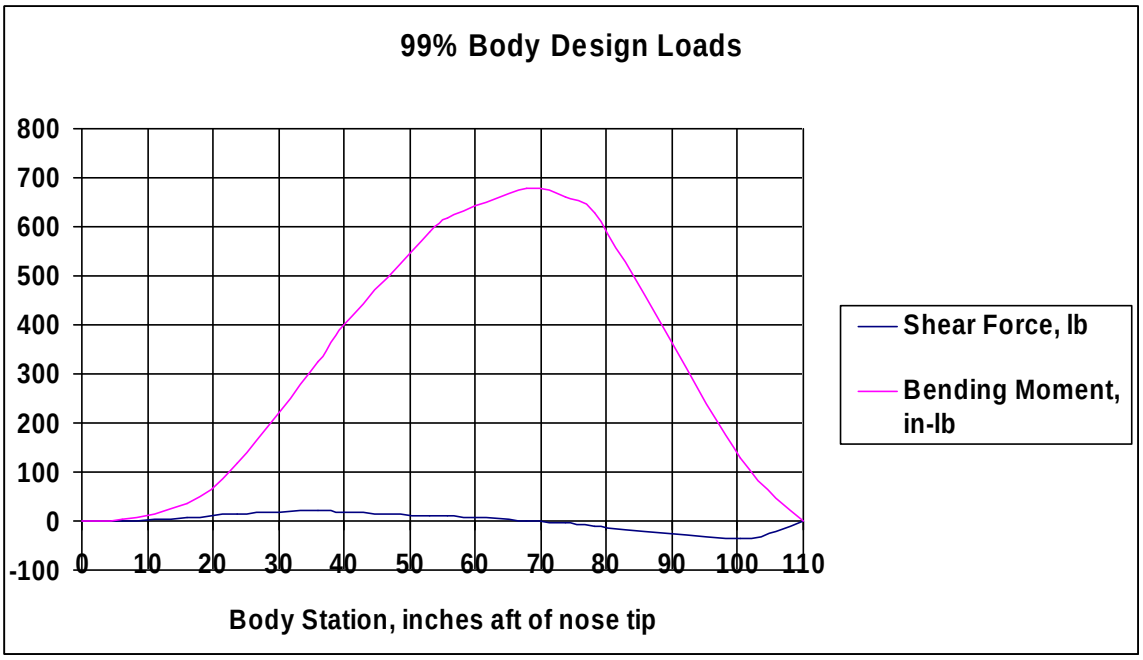
from eq. (15) for M_B into eq. (17) taking σ to be the standard deviation in planar bending moment. An analogous analysis for an asymmetric structure is described in Appendix G.

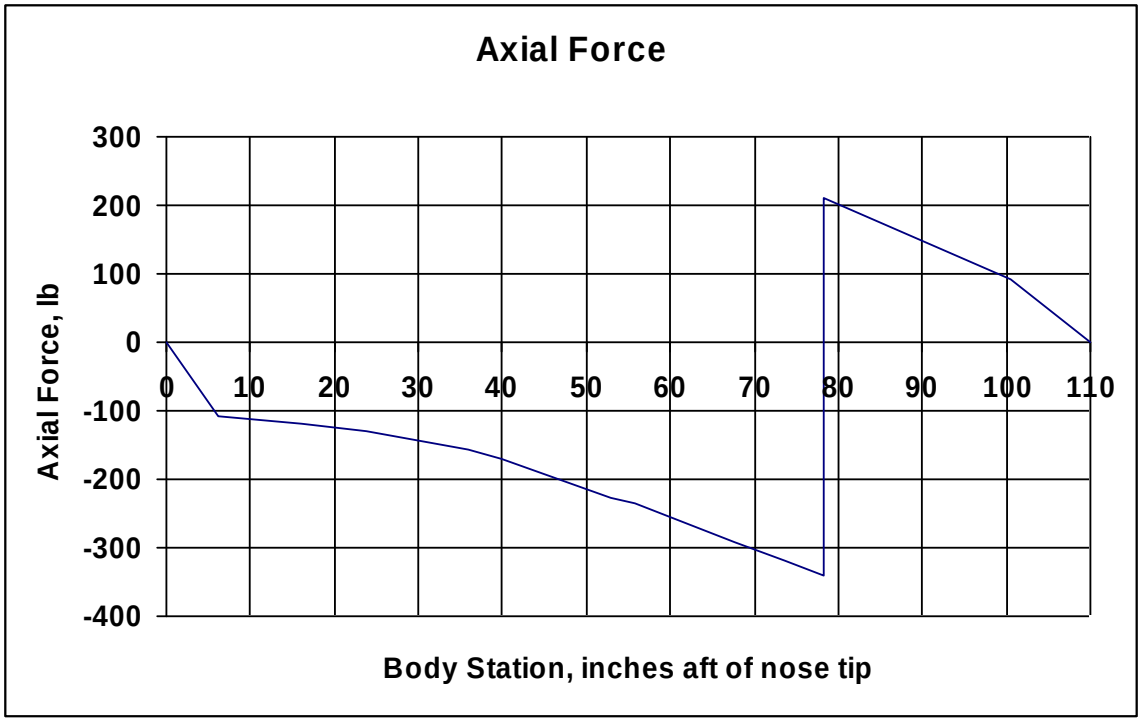
Fin loading statistics are different because fins have planar geometry. Given the mean and standard deviation in fin bending moment profile, the one dimensional cumulative Gaussian distribution will generate the probability that any given bending moment, $Lfin$, will not be exceeded. Fortunately, Excel provides this function so BENDIT7 can provide consistent structural loads for both rocket body and fins.

Summary

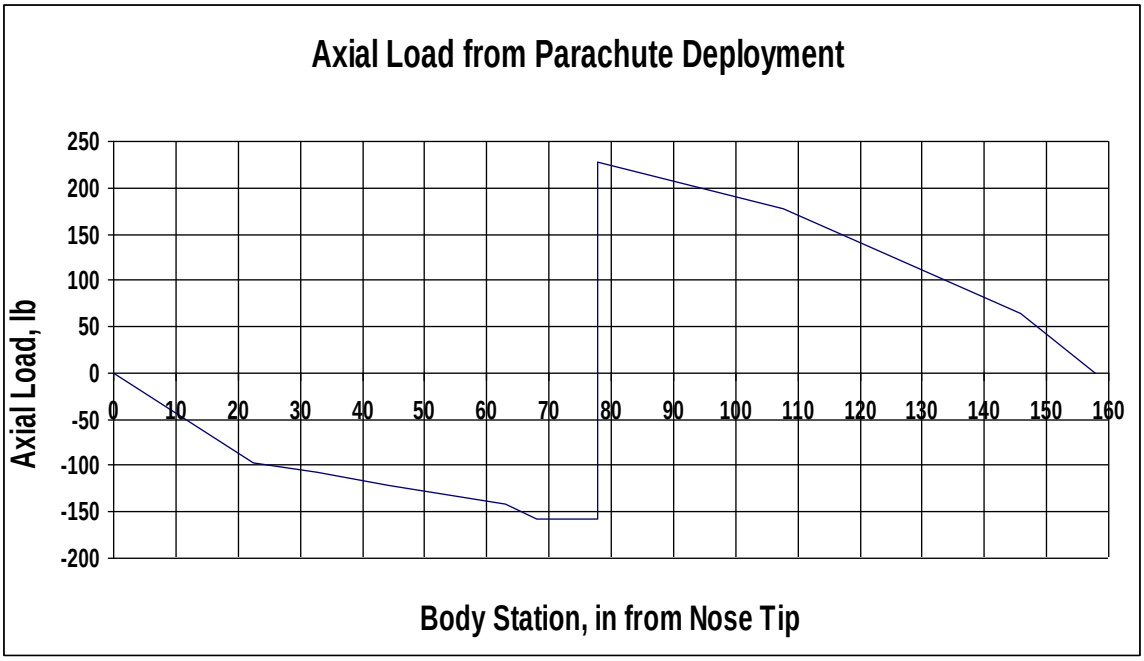
This report documents the BENDIT7 code that generates design structural loads for rocket bodies and fins. These results are prepared for an input value of the probability that these loads will not be exceeded.

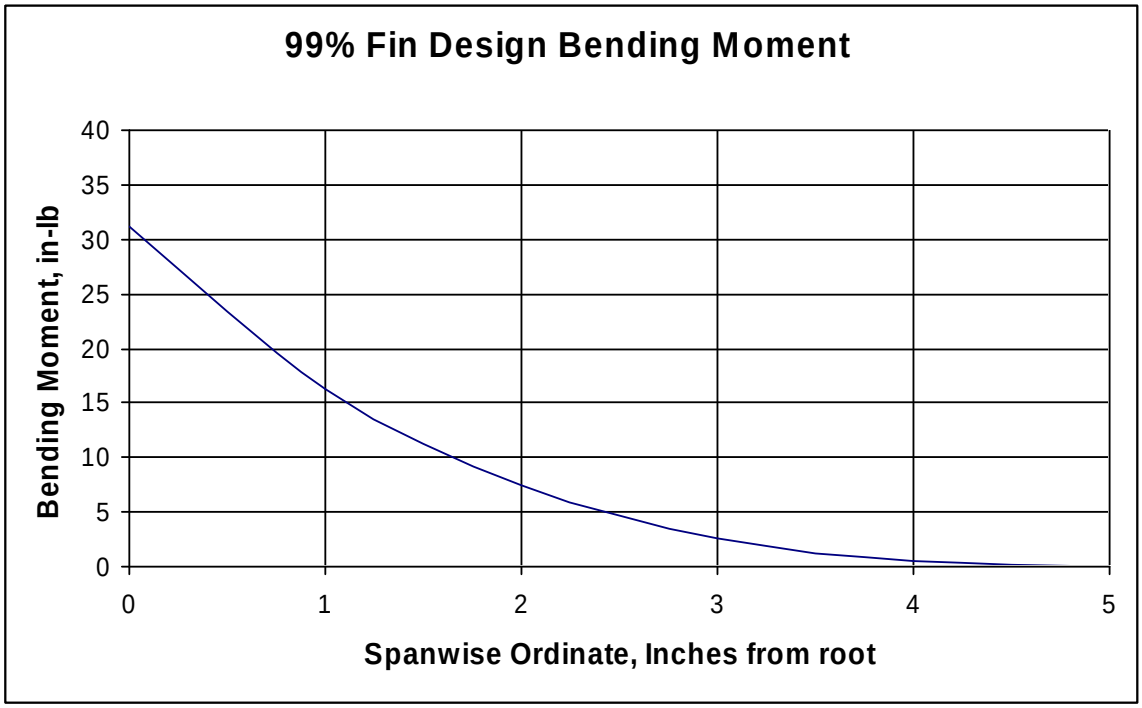
As an example, the 99% reliable structural loads for an M class ESRA-like rocket at burnout as developed from BENDIT are displayed below:





As expected, the shear force resembles the derivative of the bending moment. The axial force shows the expected saw wave shape. It was estimated neglecting aerodynamic drag.





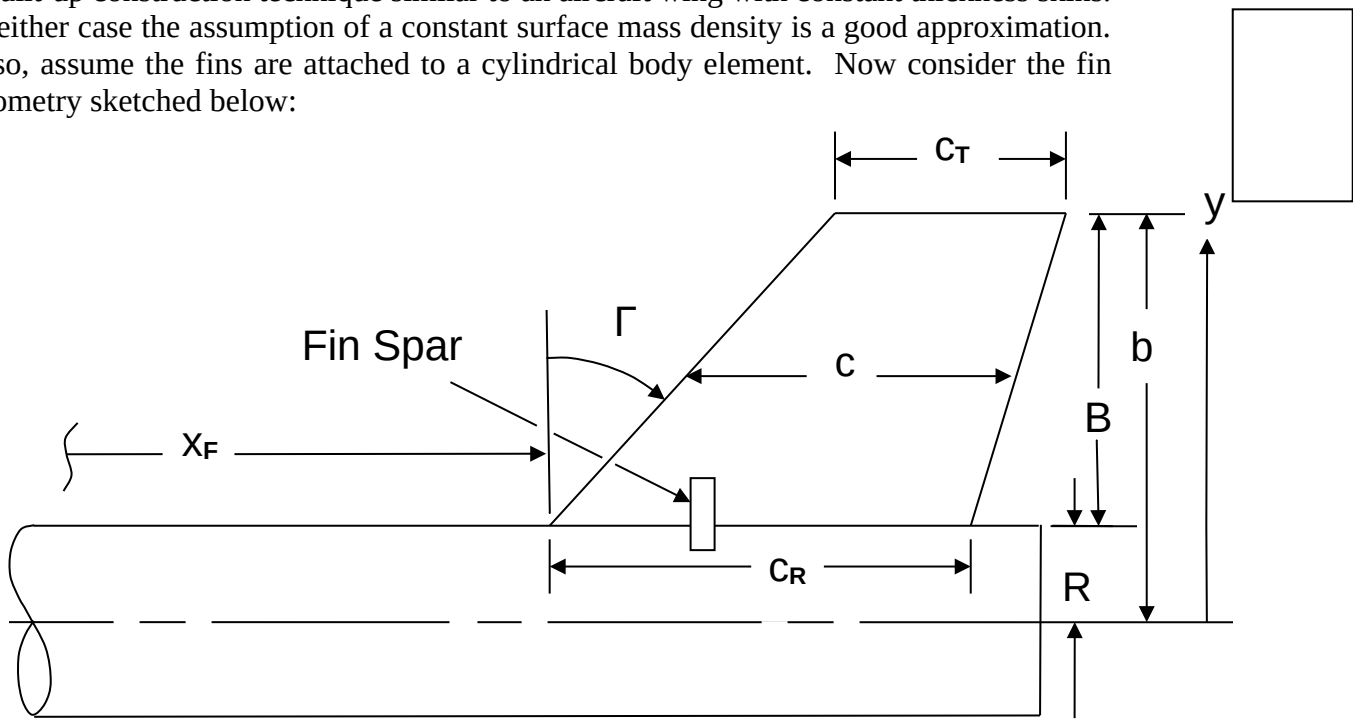
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Appendix A: Fin Mass Properties

Most small fins are constructed from a solid slab or sheet of material. Most large fins use a built-up construction technique similar to an aircraft wing with constant thickness skins. In either case the assumption of a constant surface mass density is a good approximation. Also, assume the fins are attached to a cylindrical body element. Now consider the fin geometry sketched below:



Fin Geometry

Now, suppose good design practice has been followed, and each fin has been attached to the body by its spar. All bending and shear loads are carried out via the spar; torsion is

carried out by the spar and the fin cant angle adjustment tab. It follows that the body element with the attached fin spars is the one whose mass properties must be found. Then,

Fin mass $= \mu *$ area $= \mu B(c_R + c_T)/2$, and
Total fin mass $= N\mu B(c_R + c_T)/2$.
Then since $(b - R)c(y) = A - Cy = c_R b - c_T R - (c_R - c_T)y$,

$$\text{Fin } y \text{ mass moment} = \int_0^B \mu c dy * y = \mu \int_0^B (c_R y - (c_R - c_T)y^2/B)dy$$
$$= \mu[c_R B^2/2 - (c_R - c_T)B^2/3] = \mu B^2(c_R/6 + c_T/3)$$

$$\text{Fin } x \text{ mass moment} = \int_0^B \mu c dy * x = \mu \int_0^B c dy * (y\tan\Gamma + c/2)$$
$$= \mu \int_0^B [y\tan\Gamma + (A - Cy)/2](A - Cy)dy$$
$$= \mu \int_0^B [(A^2 - 2ACy + C^2y^2)/2 + A\tan\Gamma y - C\tan\Gamma y^2]dy$$

$$= \mu[(A^2B - ACB^2 + C^2B^3/3)/2 + AB^2\tan\Gamma/2 - B^3C\tan\Gamma/3]$$

$$= \mu[(Bc_R^2 - c_R B(c_R - c_T) + (c_R - c_T)^2 B/3)/2 + c_R B^2\tan\Gamma/2 - B^2(c_R - c_T)\tan\Gamma/3]$$

These two moments place the fin center of mass (xCM, yCM) relative to the fin root chord leading edge:

$$xCM = 2[(c_R^2 - c_R(c_R - c_T) + (c_R - c_T)^2/3)/2 + c_R B\tan\Gamma/2 - B(c_R - c_T)\tan\Gamma/3]/(c_R + c_T),$$
and

$$yCM = 2B(c_R/6 + c_T/3)/(c_R + c_T).$$

The element center of mass

$$= [m_i l_i k_i + N\mu B(c_R + c_T)/2 + (xCM + (x_F - x_i))]/[m_i + N\mu B(c_R + c_T)/2] \qquad (18a)$$
from the front of the fin bearing element.

The fin contributions to the moment of inertia depend on whether the bending moment vector lies in the plane of a fin panel or is normal to it.

First, analyze the case of the bending moment vector in the plane of the fin panel, J_0. Evaluate the moment of inertia about an axis parallel to the bending moment and passing through the root chord leading edge, making frequent use of the parallel axis theorem:

$J_0 = \mu \int_0^B c((y\tan\Gamma + c/2)^2 + c^2/12)dy$, where y = 0 corresponds to the outer body surface. Then,

$$J_0 = \mu \int_0^B c(y^2\tan^2\Gamma + cy\tan\Gamma + c^2/3)dy, \text{ or}$$

$$= \mu \int_0^B (A - Cy)(y^2\tan^2\Gamma + (A - Cy)y\tan\Gamma + (A - Cy)^2/3)dy$$

$$= \mu \int_0^B [A\tan^2\Gamma y^2 + A^2\tan\Gamma y - AC\tan\Gamma y^2 - C\tan^2\Gamma y^3 - AC\tan\Gamma y^2 + C^2\tan\Gamma y^3 + A^3/3 - A^2Cy$$
$$+ AC^2y^2 - C^3y^3/3]dy, or$$
$$= \mu \int_0^B [A\tan^2\Gamma y^2 + A^2\tan\Gamma y - 2AC\tan\Gamma y^2 - C\tan^2\Gamma y^3 + C^2\tan\Gamma y^3 + A^3/3 - A^2Cy + AC^2y^2 - C^3y^3]dy$$
$$= \mu[AB^3\tan^2\Gamma/3 + A^2B^2\tan\Gamma/2 - 2ACB^3\tan\Gamma/3 - CB^4\tan^2\Gamma/4 + C^2B^4\tan\Gamma/4 + A^3B/3 - A^2CB^2/2$$
$$+ AC^2B^3/3 - C^3B^4/12]$$
$$= \mu[c_RB^3\tan^2\Gamma/3 + c_R^2B^2\tan\Gamma/2 - 2c_R(c_R - c_T)B^2\tan\Gamma/3 - (c_R - c_T)B^3\tan^2\Gamma/4 + (c_R - c_T)^2B^2\tan\Gamma/4$$
$$+ c_R^3B/3 - c_R^2(c_R - c_T)B/2 + c_R(c_R - c_T)^2B/3 - (c_R - c_T)^3B/12], or$$
$$J_0 = \mu B[(c_R + 3c_T)B^2\tan^2\Gamma + B\tan\Gamma(c_R^2 + 2c_Rc_T + 3c_T^2) + c_R^3 + c_R^2c_T + c_Rc_T^2 + c_T^3]/12 \quad (18b)$$

Next, analyze the case of the bending moment vector normal to the fin plane, J_1. Evaluate the moment of inertia about an axis parallel to the bending moment and passing through the root chord leading edge:

$$J_1 = \mu \int_0^B [c^3/12 + c(y^2 + (y\tan\Gamma + c/2)^2)]dy, or$$

$$J_1 = \mu \int_0^B [(A - Cy)^3/12 + (A - Cy)(y^2 + y^2\tan^2\Gamma + (A - Cy)y\tan\Gamma + (A - Cy)^2/4)]dy$$

$$= \mu \int_0^B [(A^3 - 3A^2Cy + 3AC^2y^2 - C^3y^3)/12 + (A - Cy)y^2(1 + \tan^2\Gamma) + (A^2 - 2ACy + C^2y^2)y\tan\Gamma$$
$$+ (A^3 - 3A^2Cy + 3AC^2y^2 - C^3y^3)/4]dy, or$$
$$J_1 = \mu \int_0^B [(A^3 - 3A^2Cy + 3AC^2y^2 - C^3y^3)/3 + (Ay^2 - Cy^3)\sec^2\Gamma + (A^2y - 2ACy^2 + C^2y^3)\tan\Gamma]dy, or$$

$$J_1 = \mu[(A^3B/3 - A^2CB^2/2 + AC^2B^3/3 - C^3B^4/12 + (AB^3/3 - CB^4/4)\sec^2\Gamma + (A^2B^2/2 - 2ACB^3/3 + C^2B^4/4)\tan\Gamma], or$$

$$J_1 = \mu[(c_R^3B/3 - c_R^2(c_R - c_T)B/2 + c_R(c_R - c_T)^2B/3 - (c_R - c_T)^3B/12 + (c_RB^3/3 - (c_R - c_T)B^3/4)\sec^2\Gamma + (c_R^2B^2/2 - 2c_R(c_R - c_T)B^2/3 + (c_R - c_T)^2B^2/4)\tan\Gamma], or$$

$$J_1 = \mu B[(c_R^3/3 - c_R^2(c_R - c_T)/2 + c_R(c_R - c_T)^2/3 - (c_R - c_T)^3/12 + (c_RB^2/3 - (c_R - c_T)B^2/4)\sec^2\Gamma + (c_R^2B/2 - 2c_R(c_R - c_T)B/3 + (c_R - c_T)^2B/4)\tan\Gamma]/12, or$$

$$J_1 = \mu B[(c_R^3 + c_R^2c_T + c_Rc_T^2 + c_T^3) + B^2(c_R + 3c_T)\sec^2\Gamma + B(c_R^2 + 2c_Rc_T + 3c_T^2)\tan\Gamma]/12 \quad (18c)$$

These results are directly applicable to a four-finned rocket, the most common configuration. The same analysis technique can be applied to other numbers of fins, but, there is some additional labor involved.

Appendix B: Conical Element Mass Properties

It's more than a little likely that some of the body elements will be conical. As in the main body of this memo, it is assumed that any conical element mass is obtained offline and provided to BENDIT7 as an input. However the element center of mass is probably not readily available offline, and therefore must be generated within BENDIT7. Assuming such an element has a constant volumetric mass density, the center of mass as a fraction of the element length l_i, as measured from the larger end, k_i, can be shown to be

$$k_I = (1 + 2z + 3z^2)/(4(1 + z + z^2)),$$

where z = Smaller radius/Larger radius

An appropriate reference for this result is William H. Beyer, editor, "CRC Standard Mathematical Tables", 27th Edition, CRC Press, Inc., page 131.

While a purist might demand the element pitch moment of inertia be also amended to reflect the conical geometry, the constant volumetric mass density assumption becomes significantly shakier. An analyst who wishes to capture this effect more accurately has only to break the body up into shorter elements.

Appendix C: Fin Inertial Relief Bending Moment

There will also be an inertial reaction to the imposed body angle of attack[6], and it will act in a way that opposes the aerodynamic loads given by eq. (10a).

[6] The inertial reactions to δ_F and λ_R loadings are much smaller than those due to α, and are therefore neglected.

$$MI(y) = -\int_{y}^{b} \mu c(\eta)(\eta - y)\left[(d^2\alpha/dt^2)(x_F + xCM - X) + A_z\right]d\eta$$

where $MI(y) =$ The inertial pitch reaction fin bending moment. Then

$$MI(y) = -\mu\left[(d^2\alpha/dt^2)(x_F + xCM - X) + A_z\right]\left[A(b^2 - 2by + y^2)/2 + C(-2b^3 + 3b^2y - y^3)/6\right]$$

Next, use eq. (6) to find $d^2\alpha/dt^2$:

$$MI(y) = -\mu\left[\omega_n^2(x_F + xCM - X) + A_z/\alpha\right]\left[A(b^2 - 2by + y^2)/2 + C(-2b^3 + 3b^2y - y^3)/6\right]\alpha \quad (19a)$$

Finally, since the mean angle of attack vanishes, the mean inertial relief bending moment also vanishes. When this is added back into eq. (11a), the function $F(y)$ must be redefined as $F_{TOT}(y)$:

$$F_{TOT}(y) = F(y) - \mu\left[\omega_n^2(x_F + xCM - X) + A_z/\alpha\right]\left[A(b^2 - 2by + y^2)/2 + C(-2b^3 + 3b^2y - y^3)/6\right] \quad (19b)$$

Using eq. (19b) in eq. (12a) will provide the appropriate fin bending moment result. In BENDIT7 this result is coded to find the fin bending moment standard deviation.

Appendix D: Approximate Angle of Attack Estimates

First, consider a single stage rocket in boost for which the design loads can be expected around burnout. Other flight conditions such as upper stage ignition will require slightly different treatments.

References (1) and (2) provide formal estimates of the angles of attack due to gust, and thrust and fin misalignment that are both dynamically and statistically valid. These are developed in terms of pitch and roll wavenumbers. See ref. (4) for a description of the pitch wavenumber. Basically, a wavenumber is just a frequency in the altitude domain rather than the more familiar frequency in the time domain.

The gust model presented here assumes that the rocket encounters a stochastic gust field described by a Dryden autocorrelation function. Then, neglecting any short period damping, eq. (13) of ref. (2) can be used:

The gust model presented here assumes that the rocket encounters a stochastic gust field described by a Dryden autocorrelation function. Then, neglecting any short period damping, eq. (13) of ref. (2) can be used:

$$\mathrm{var}(V_z(h)) = \frac{\lambda_P^2\sigma_G^2}{(\lambda_G^{*2} + \lambda_P^2)}\left[\begin{array}{l}\lambda_G^*(h - L) - \lambda_G^*\sin(2\lambda_P(h - L))/2\lambda_P - \sin^2(\lambda_P(h - lL)) \\ + \frac{2\lambda_P^2}{(\lambda_G^{*2} + \lambda_P^2)} - \frac{2\lambda_P}{(\lambda_G^{*2} + \lambda_P^2)}\exp(-\lambda_G^*(h - L))\left(\begin{array}{l}\lambda_G^*\sin(\lambda_P(h - L)) \\ + \lambda_P\cos(\lambda_P(h - L))\end{array}\right)\end{array}\right] \quad (20)$$

Then, the 1 sigma, 1 plane gust angle of attack is

$$\sigma_{G\alpha} = \sqrt{\text{var}(V_z(h))}\,/\,U \tag{21}$$

For events after first stage burnout, the asymptotic bound of eq. (17), as found in ref. (2), can be used

$$\sigma_{G\alpha} = \sqrt{\frac{\lambda_P^2 \lambda_G^* \sigma_G^2}{2a(\lambda_P^2 + \lambda_G^{*2})}}$$

Gust parameter data can be found in ref. (8).
Reference (1) includes steady state angle of attack estimates for thrust misalignment, fin misalignments and C.G. offset for a rolling rocket. The results are that the 1 sigma, 1 plane steady state thrust misalignment response is

$$\sigma_{T\alpha} = T \frac{\sqrt{(\text{var}(l\delta_T) + \text{var}(\Omega))}}{I_{YY} U^2 (\lambda_P^2 - \lambda_R^2)} \tag{22a}$$

This is valid except immediately after ignition of an upper stage. The transient response after upper stage ignition includes an overshoot the doubles the angle of attack. That is, immediately after upper stage ignition

$$\sigma_{T\alpha} = 2T \frac{\sqrt{(\text{var}(l\delta_T) + \text{var}(\Omega)}}{I_{YY} U^2 (\lambda_P^2 - \lambda_R^2)} \tag{22b}$$

See references (6) and (7) for parameter data on δ_T and Ω.

As always, because these two angle variances are statistically independent, the system variance is the sum of its two contributors. The steady state fin misalignment response is that:

$$\sigma_{F\alpha} = \sqrt{\frac{N}{8}} \rho S_{ref} d_{ref} C_{M\delta F} \sigma_{\delta F} / I_{YY} (\lambda_P^2 - \lambda_R^2)$$

(23)

Again, ref. (6) has data for fin misalignment errors. The stability derivative, $C_{M\delta F}$, can be obtained from the fin assembly normal slope due to cant angle and its center of pressure from the BARROWMAN EQUATIONS, ref's. (9), (17) & (18), and knowledge of the rocket center of gravity. Finally, the complete system planar σ_α is just the RSS of the four independent contributors

$$\sigma_\alpha = \sqrt{\sigma_{G\alpha}^2 + \sigma_{T\alpha}^2 + \sigma_{F\alpha}^2} \tag{24}$$

A summary of typical results is shown below. These results are for a typical single stage M class sounding rocket flown in the ESRA-sponsored IREC competition evaluated at maximum velocity (near burnout):

Summary of Typical Angle of Attack Contributions

Source	1 σ Alpha, rad
Thrust Misalignment	0.044730143
Fin Misalignment	0.001774548
Center of Gravity Offset	0.006957534
Gusts	0.020323395
Root Sum Square	0.049652617

The large effect due to thrust misalignment, typical of small sounding rockets, strongly suggests that coasting flight need not be considered for evaluating design flight loads.

Appendix E: Roll Rate Selection

For multistage rockets, first establish the last stage fin cant angle, then the penultimate stage fin cant angle, and so on. There are three possible choices for selecting the booster roll rate and fin cant angle. Only two of these apply to upper stage(s). For a single stage rocket the mean fin cant angle $\bar{\delta}_F$ for a given λ_R can be found from ref. (20):

$$\lambda_R = - \frac{C_{L\delta} N \bar{\delta}_F}{R C_{Lp}} \left[\frac{2\pi\rho(h - L)R^4 C_{Lp}}{2\pi\rho(h - L)R^4 C_{Lp} - I_{xx}} \right] \quad . \tag{25a}$$

Equation (23a) also applies to a two stage configuration if

$C_{Lp} = \sum C_{Lp}$ and $C_{L\delta}\bar{\delta}_F = \sum C_{L\delta}\delta_F$ where the summations extend over all stages.

The first choice for the booster $\bar{\delta}_F$ is to set the total number of roll revolutions to an integer value to minimize trajectory dispersions arising from body-fixed perturbations. Trajectory dispersion considerations are relatively far less important for the upper stages, and, so this does not apply to them. See eq. (13a) on page 15 for the details. Then, the ideal the boost phase roll angle at booster burnout would be an integer number M of roll cycles. That is,

$$\Delta\phi_{BO} = 2\pi M = - \frac{C_{L\delta} N \bar{\delta}_F}{R C_{LP}} \left[h - L + k \log\left(\frac{h - L - k}{- k} \right) \right], \tag{25b}$$

where $$k = \frac{I_{XX}}{2\pi\rho R^4 C_{Lp}}. \qquad (25c)$$

Again, this result also applies to a two stage rocket if the codicils developed for eq. (23a) above apply. M should be at least 2 or 3. More is better. Once M has been selected, eq's. (25b) and (25c) can be used to establish the mean fin cant angle $\bar{\delta}_F$. The corresponding roll wavenumber λ_R is given by eq. (25a).

The second possible choice, applicable to any stage, is to select λ_R to be an inputted fraction of the pitch wavenumber, λ_P. There are two conflicting considerations. First, the body-fixed perturbations elicit an angle of attack (and structural loads) response proportional to $\frac{\lambda_P^2}{\lambda_P^2 - \lambda_R^2}$. Thus, roll rate tends to "destroy" the pitch static stability. However, as long as λ_R is less than 30-40% of λ_P the damage is not too great. Then for a single stage rocket the roll wavenumber λ_R, as input can be used to find the mean fin cant angle $\bar{\delta}_F$ from eq. (25a).

When analyzing loads on an upper stage at its ignition, the high dynamic pressure usually occurring there implies the λ_R roll transient from the lower to upper stage will be very rapid. Assume it is instantaneous. Note that both λ_P and λ_R will be proportional to flight speed. It follows that λ_R as a fraction of λ_P at ignition will be the fraction near second stage burnout. Then, the upper stage roll wavenumber at ignition is just the steady state asymptotic value obtained from eq. (25a),

$$\lambda_R = - \frac{C_{L\delta} N \bar{\delta}_F}{R C_{Lp}}, \qquad (25d)$$

where $\bar{\delta}_F$ is the mean upper stage cant angle.

The third choice is to do nothing to adjust fin cant angles. When fin cant angles for small rockets are calculated it often turns out that these angles are to small too be accurately measured without special equipment. Under these conditions one frequently decides to fly with the so called "natural roll rate", the roll rate obtained from assembly tolerance build up with no attempt made to apply corrections. Equation (13a) on page 15 of this is our starting point. For a single stage rocket, assuming all fins have the same cant angle statistics, this can be manipulated to give the variance in λ_R:

$$\mathrm{var}\,\lambda_R = \frac{N\left(\sum_{\text{fins}} C_{L\alpha}\right)^2 \mathrm{var}\,\delta_F}{R^2\left(\sum_{\text{fins}} C_{L\alpha}\right)^2}\left[\frac{1}{1 - \frac{I_{XX}}{2\pi\rho(h-1)R^4 C_{Lp}}}\right]^2. \qquad (26a)$$

Now, by appealing to the central limit theorem, it can be safely assumed that δ_F has a normal distribution. Therefore, so does λ_R. But, only the absolute value of λ_R appears in

every context in which it occurs. The distribution for the absolute value of a normal variable is just a normal distribution folded about its ordinate. Therefore, assuming no bias errors, it is easy to show that the mean value of $|\lambda_R|$ is given by

$$mean\,|\,\lambda_R\,|=\sqrt{\frac{2\,\mathrm{var}\,\lambda_R}{\pi}} \tag{27a}$$

The corresponding fin cant angle is

$$mean\,|\,\delta_F\,|=\sqrt{\frac{2N}{\pi}}\sigma_{\delta F}\,. \tag{27b}$$

For upper stages the cant angles are all found from eq. (27b). The variance in $|\,\lambda_R\,|$ is found from the steady state version of eq. (26a):

$$\mathrm{var}\,\lambda_R=\frac{N\{\sum_{stages} C_{L,W}\}^2\,\mathrm{var}\,\delta_F{}^2}{R^2\{\sum_{stages} C_{L,P}\}^2} \tag{26b}$$

When applying these results, one must start with the idea that any given rocket can have only one roll program, and this will be manifested in all loading conditions analyzed. We consider only three choices for the first stage fins:

1. Roll a specified integer number of turns from liftoff to burnout
2. Set λ_R to a specified fraction of the burnout λ_P
3. Take what ever the shop gives us, i.e., use a random roll program

Denote the chosen roll program by a digraph (1st, 2nd). For example, roll program (1,3) tells us that the first stage fins will be set to yield a specified number of turns during boost while the second stage fins will not be adjusted at all. Or, (3,3) implies both stages will roll randomly. Since choice 1. only applies to the first stage, there will be only 6 possible roll programs. Since there are only three main loading conditions identified on the bottom of page 7, the following table of fin cant angles and roll wavenumbers can be prepared. It was predicated on the assumption that there will always be at least a brief coast between first stage burnout and second stage ignition. In the following table, when more than one stage is analyzed as a system, one always finds the upper stage fin cant angle, and use these in estimating the booster roll wavenumber and cant angles. The table is provided as a guide to coding these roll models into BENDIT7.

Summary of Roll Models

		Stage 1	Stage 1	Stage 2	Stage 2
Analysis Condition	Roll Digraph	λ_R	δ_F	λ_R	δ_F
1st Stage BO	(1,2)	Eq. (25a) +	Eq. (25b) +	% λ_p 2nd stage	Eq. (25a)
1st Stage BO	(1.3)	Eq. (25a) +	Eq. (25b) +	Eq. (27a)	Eq. (27b)
1st Stage BO	(2,2)	% λ_p 1st stage	Eq. (25a)	% λ_p 2nd stage	Eq. (25d)
1st Stage BO	(2.3)	% λ_p 1st stage	Eq. (25a)	Eq. (27a)	Eq. (27b)
1st Stage BO	(3,2)	Eq. (26a)	Eq. (27b)	% λ_p 2nd stage	Eq. (25d)
1st Stage BO	(3,3)	Eq's. (26a) & (27a)	Eq. (27b)	Eq. (27a)	Eq. (27b)
2nd Stage IG	(1,2)	NA	NA	% λ_p 2nd stage	Eq. (25d)
2nd Stage IG	(1.3)	NA	NA	Eq. (27a)	Eq. (27b)
2nd Stage IG	(2,2)	NA	NA	% λ_p 2nd stage	Eq. (25d)
2nd Stage IG	(2.3)	NA	NA	Eq. (27a)	Eq. (27b)
2nd Stage IG	(3,2)	NA	NA	% λ_p 2nd stage	Eq. (25d)
2nd Stage IG	(3,3)	NA	NA	Eq. (27a)	Eq. (27b)
2nd Stage BO	(1,2)	NA	NA	% λ_p 2nd stage	Eq. (25d)
2nd Stage BO	(1.3)	NA	NA	Eq. (27a)	Eq. (27b)
2nd Stage BO	(2,2)	NA	NA	% λ_p 2nd stage	Eq. (25d)
2nd Stage BO	(2.3)	NA	NA	Eq. (27a)	Eq. (27b)
2nd Stage BO	(3,2)	NA	NA	% λ_p 2nd stage	Eq. (25d)
2nd Stage BO	(3,3)	NA	NA	Eq. (27a)	Eq. (27b)

Appendix F: Roll Stability Derivatives

Reference (3) documents strip theory for setting fin cant angles. As a by-product, it also develops the roll stability derivatives, for the single fin driving and overall damping torques:

$$C_L = L_{fin} / qS_{ref} d = L_{fin} / 2\pi R^3 q \qquad (28a)$$

Then,

$$C_{L\delta} = \frac{\partial C_{Lfin}}{\partial \delta_F} = c_{N\alpha}(b - R)\left[A(b+R)/2 - C(b^2 + bR + R^2)/3\right]/2\pi R^3, \qquad (28b)$$

and

$$C_{Lp} = \frac{\partial C_L}{\partial(\frac{\omega_R d}{2U})} = \frac{\partial C_L}{\partial(\frac{\omega_R R}{U})}, \text{ or}$$

(28c)

$$C_{Lp} = c_{N\alpha}(b - R)N\left[-A(b^2 + bR + R^2)/3 + C(b^3 + b^2 R + bR^2 + R^3)/4\right]/2\pi R^4 \qquad (28d)$$

Here, A and C are defined on page 11. Both these derivatives are based on body cross section area and body diameter. The damping derivative is taken with respect to $\omega_R R/U$.

Finally, the airfoil normal force slope $c_{N\alpha}$ with respect to local angle of attack is an average across the fin span. Take it to be just that for an isolated single fin panel, referenced to its own area, from the relevant BARROWMAN EQUATIONS:

$$c_{N\alpha} = C_{N\alpha} \qquad (29)$$

This implies that the "average airfoil" normal force coefficient slope includes the effects of tip losses, but no body interference. We have already included an inverse square body upwash in the loading so that this will cover all the loads on the exposed fin panels. Note that the entire fin assembly normal force coefficient slope, an input to BENDIT7, includes the fin alone and the body-on-fin interference and the fin-on-body interference per ref. (5).

In the case of a multistage rocket, one just adds the individual C_{Lp}s to arrive at the system level result. The same should be done for the stage $C_{L\delta}\delta_F$s to obtain a system level result.

Appendix G: Pitch/Yaw Aerodynamics

The pitch/yaw aerodynamic data must be input from another analysis tool(s) in the form of normal forces and local centers of pressure. There are two such that can be used in support of BENDIT7, references (17) and (18), for subsonic and supersonic flight respectively. But, pitch moment and pitch dynamics data require knowledge of the overall center of mass and moment of inertia. BENDIT7 generates these internally as described by eq's. (2a) and (2b).

Each set of fins can generate aerodynamic normal forces consisting of loading on the fins themselves and loading carried over to the body of revolution. The loads on the fins themselves include interference effects due to the body. Also, each type of aerodynamic load has its own distinct center of pressure which can lead to the aerodynamic load being spread over two body segments. Finally, it is assumed that the centers of pressure due to angle of attack are the same as those due to fin cant angle.

Then, following ref. (5), the fin contribution(s) to a single stage configuration can be described as

$$dC_{M\alpha} = (X - x_{CPF\alpha})K_{W(B)}C_{N\alpha F}(N/2) + (X - x_{CPB\alpha})K_{B(W)}C_{N\alpha F}(N/2),$$

and (30)

$$dC_{M\delta F} = (X - x_{CPF\alpha})k_{W(B)}C_{N\alpha F}) + (X - x_{CPB\alpha})k_{B(W)}C_{N\alpha F}.$$

Here the first term in each equation arises from the loads on the fin panels and the second from the body carryover loads. For a two stage configuration, each stage contributes all the terms in eq's. (22), but with an additional vortex interference term appearing for the first stage.

$$\Delta C_{M\alpha} = (X - x_{CPF\alpha})\Delta C_{N\alpha V} \tag{31}$$

The body contributions to the normal force are documented in ref. (16) for supersonic flow, in ref. (15) for subsonic flow, and in ref. (16) for blunt supersonic regions. Results for these air loads, and their local centers of pressure, must be input from offline source(s). Their contributions to $C_{M\alpha}$ are found in the usual way.

Reference (4) provides the definition of the pitch /yaw wavenumber and pitch natural frequency:

$$\lambda_P = \sqrt{-\rho S_{ref} d_{ref} C_{M\alpha} / 2I_{YY}},$$

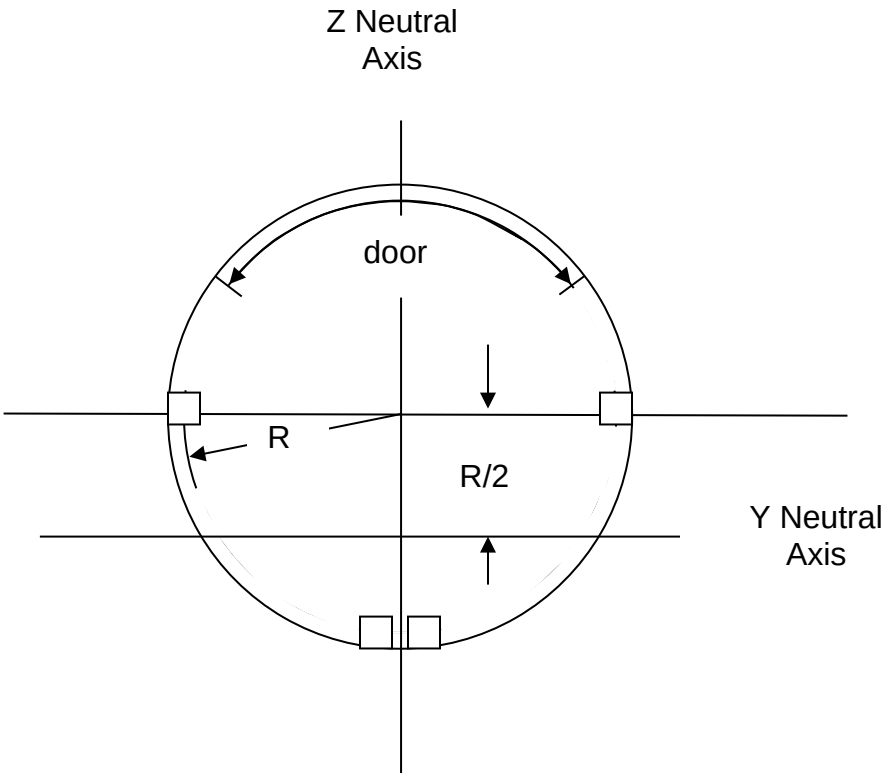
and

$$\omega_N = \lambda_P U$$

Appendix H. Loading on an Asymmetric Structure

This topic originated with a wide door needed for parachute deployment. Typically, this implies cutting away one of four longerons, and doubling up the remaining longeron in that plane.

The sketch below shows the general cross section arrangement. The longerons are shown as simple lumped areas; their detailed shape is unimportant. Simple direct calculation locates the Y neutral axis as sketched.



Cross Section Sketch of an Asymmetric Structure

There are three sources of loading: axial force, bending about the Y axis and bending about the Z axis. Now label the three structural elements according to their location on the face of a clock. That is, the right hand longeron is labeled with a subscript 3.

The static equilibrium equations are

$$P = p_3 + p_6 + p_9,$$

$$p_3 = p_9, \& \tag{32}$$

$$P = p_3 - p_6 + p_9,$$

where P = total axial load, and
p_3, p_6 & p_9 = loads carried by the individual structural elements.

These can be solved to yield

$$p_3 = p_9 = P/2, \text{and}$$

$$p_6 = 0. \tag{33}$$

The interesting thing is that the lateral members, p_3 and p_9 carry twice the axial load that they would have had all four longerons been in place.

Bending about the Z axis is especially simple:

$$M_Z = 2RP_Z, \tag{34a}$$

where M_Z =the bending moment about the Z axis, and
P_Z =the axial force carried by p_3 or p_9.

Bending about the Y axis is not much more difficult.

$$p_3 = p_9, \text{ (from symmetry)}$$

$$p_3 + p_9 = p_6 \text{ (from total force =0), and}$$

$$M_Y = (R/2)(p_3 + p_9 + p_6).$$

These can be solved to give

$$p_3 = p_9 = M_Y / 2R, \text{ and}$$
$$p_6 = M_Y / R \tag{34b}$$

When the loads from all three sources are combined,

$$p_3 = p_9 = P/2 + (M_Y + M_Z)/2R, \text{ and}$$
$$p_6 = M_Y / R \tag{34c}$$

Note that the load p_6 is spread over two adjacent longerons.

A brief example illustrates these results. Let

$R = 3''$,
$P = 480$ lbs,
$M_B = 15981$ in-lbs and
$M_D = 13286$ in-lbs.
Both bending moments are at the 99.9% level as found in BENDIT6.2. Then the worst case longeron load p is

$p = 2783.5$ lbs for the symmetrically arranged longerons, and
$p = 3371.5$ lbs for the asymmetrically arranged longerons.

Appendix I. Fin Centripetal Acceleration

Rolling fins experience a centripetal acceleration that requires a tensile load on the fin attachment point. While this is usually small, it can be significant for spin stabilized rockets. Now, consider a infinitesimal strip of chordwise fin:

$$dTL = -\mu c \omega_R^2 y dy$$

The local chord can be found from eq. (10) on page 13. Then, after integration,

$$TL = \mu \omega_R^2 \left[\frac{1}{2} \left(\frac{c_R b - c_T R}{b - R} \right) (b^2 - y^2) - \frac{1}{3} \left(\frac{c_R - c_T}{b - R} \right) (b^3 - y^3) \right]$$

The most critical part of the fin is at its root where it is attached to the body.

$$TL(R) = \mu \omega_R^2 \left[\frac{1}{2} (c_R b - c_T R)(b + R) - \frac{1}{3} (c_R - c_T)(b^2 + bR + R^2) \right] \qquad (35)$$

Equation (35) provides an estimate on the tensile load acting on the root attachment point of a spinning fin.